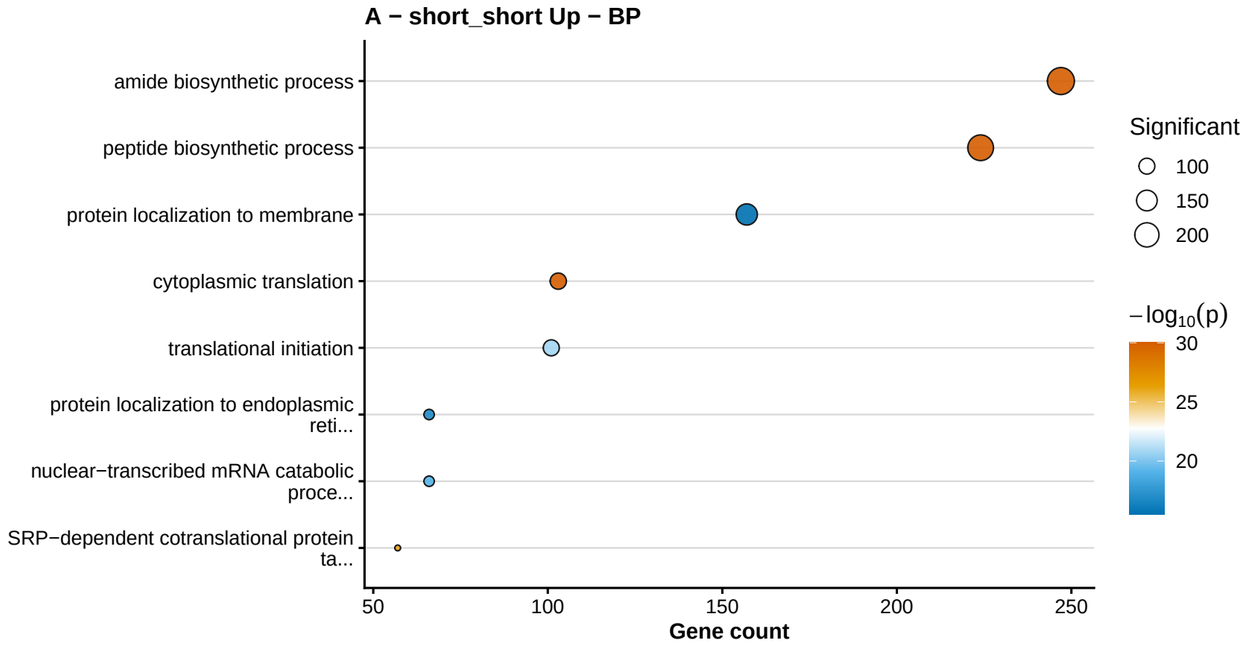
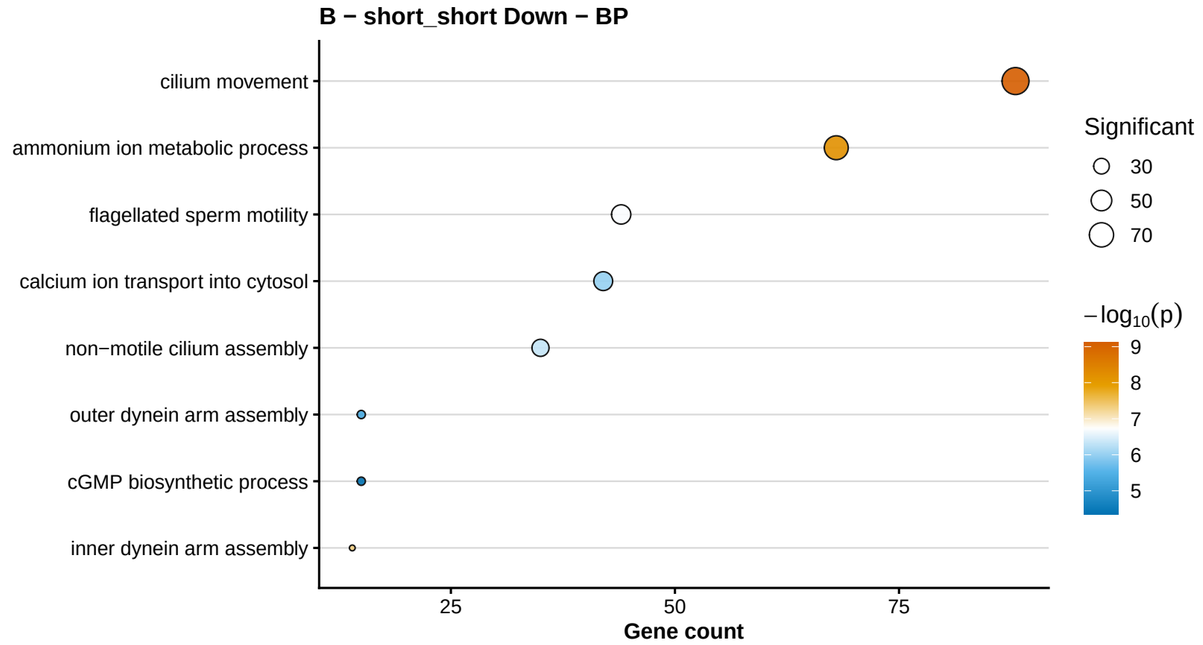
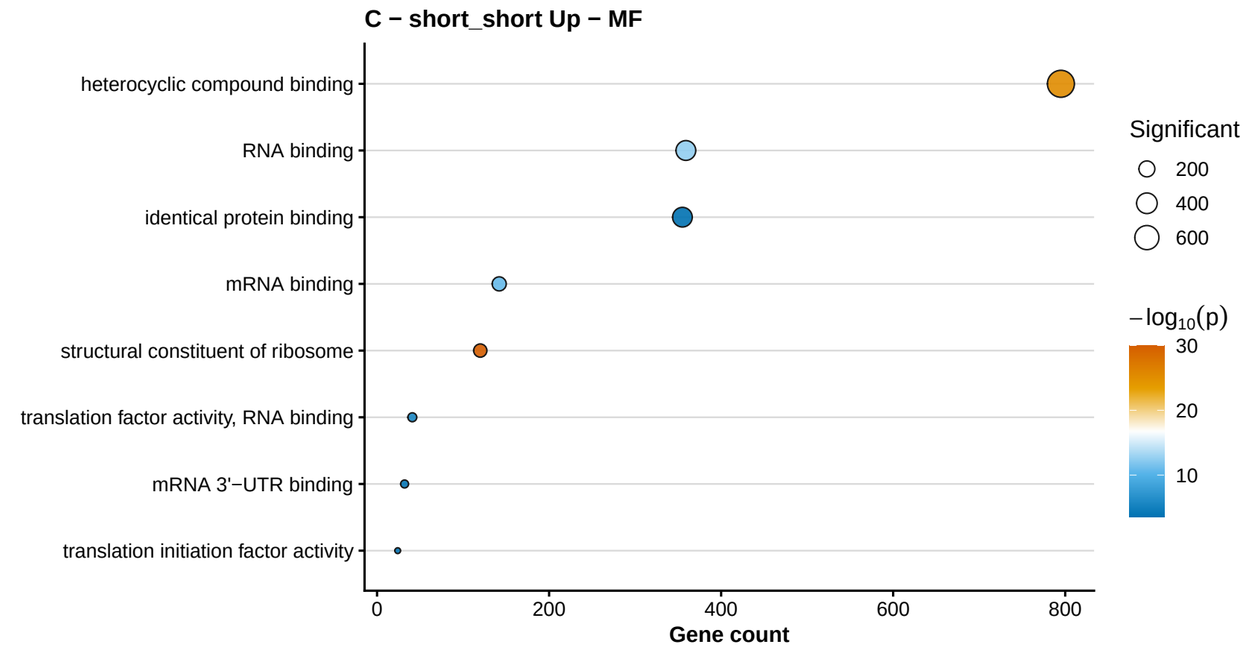
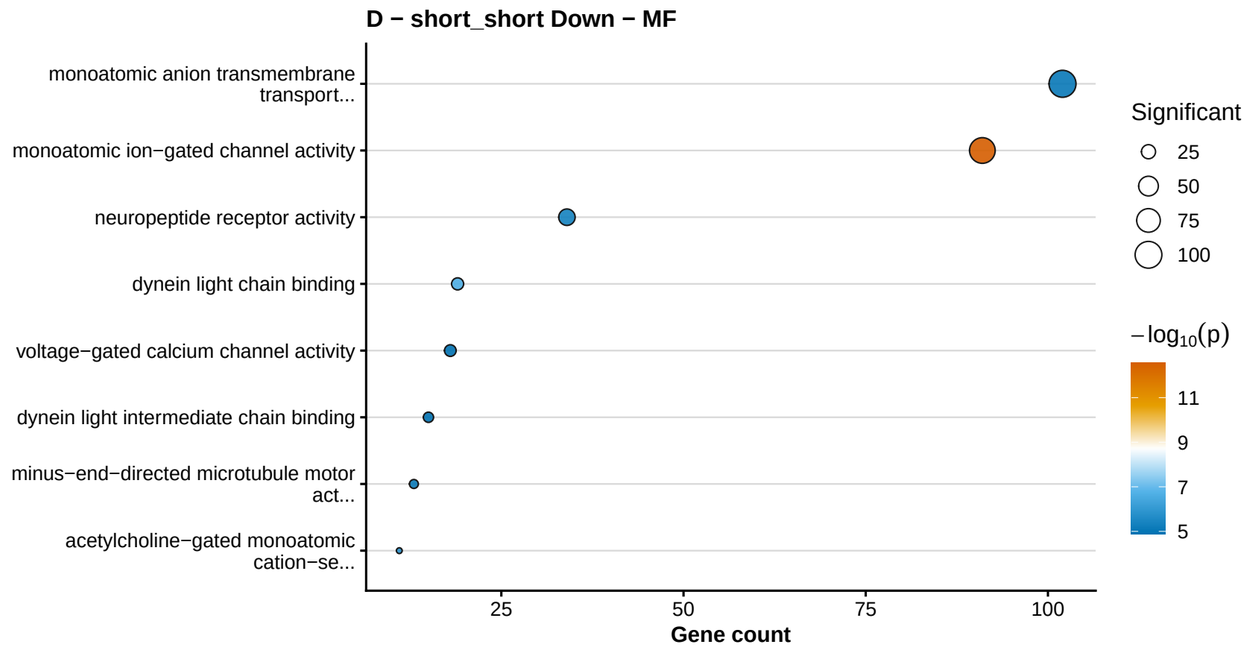
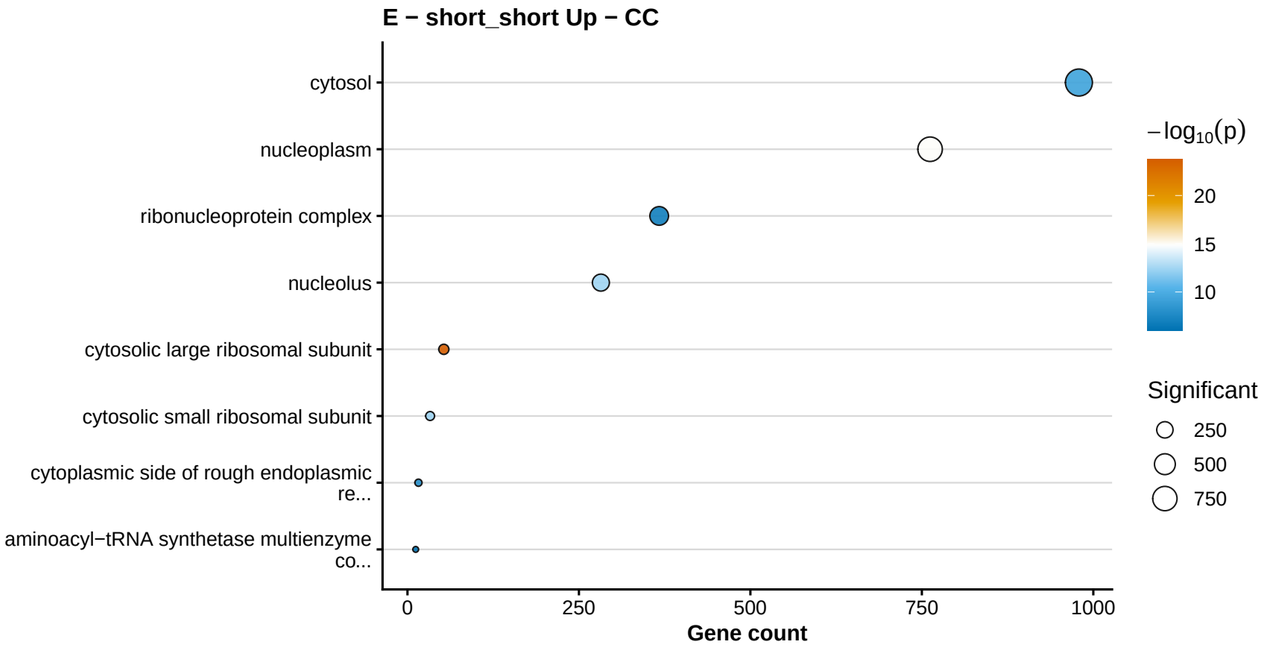
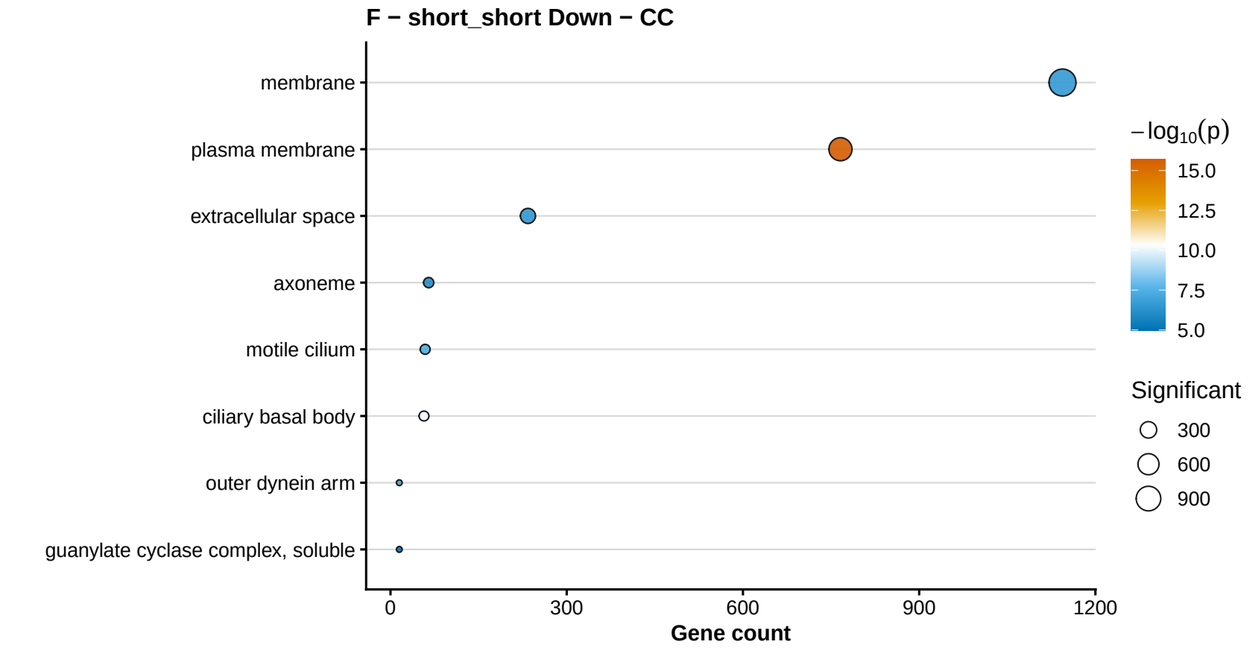

A − short_short Up − BP
amide biosynthetic process
peptide biosynthetic process
protein localization to membrane
cytoplasmic translation
translational initiation
protein localization to endoplasmic reti...
nuclear−transcribed mRNA catabolic proce...
SRP−dependent cotranslational protein ta...
50
100
150
200
250
Gene count
Significant
100
150
200
−log10(p)
30
25
20


B − short_short Down − BP
cilium movement
ammonium ion metabolic process
flagellated sperm motility
calcium ion transport into cytosol
non−motile cilium assembly
outer dynein arm assembly
cGMP biosynthetic process
inner dynein arm assembly
25
50
75
Gene count
Significant
30
50
70
−log10(p)
9
8
7
6
5


C − short_short Up − MF
heterocyclic compound binding
RNA binding
identical protein binding
mRNA binding
structural constituent of ribosome
translation factor activity, RNA binding
mRNA 3'−UTR binding
translation initiation factor activity
0
200
400
600
800
Gene count
Significant
200
400
600
−log10(p)
30
20
10


D − short_short Down − MF
monoatomic anion transmembrane transport...
monoatomic ion−gated channel activity
neuropeptide receptor activity
dynein light chain binding
voltage−gated calcium channel activity
dynein light intermediate chain binding
minus−end−directed microtubule motor act...
acetylcholine−gated monoatomic cation−se...
25
50
75
100
Gene count
Significant
25
50
75
100
−log10(p)
11
9
7
5


E − short_short Up − CC
cytosol
nucleoplasm
ribonucleoprotein complex
nucleolus
cytosolic large ribosomal subunit
cytosolic small ribosomal subunit
cytoplasmic side of rough endoplasmic re...
aminoacyl−tRNA synthetase multienzyme co...
0
250
500
750
1000
Gene count
−log10(p)
20
15
10
Significant
250
500
750


F − short_short Down − CC
membrane
plasma membrane
extracellular space
axoneme
motile cilium
ciliary basal body
outer dynein arm
guanylate cyclase complex, soluble
0
300
600
900
1200
Gene count
−log10(p)
15.0
12.5
10.0
7.5
5.0
Significant
300
600
900